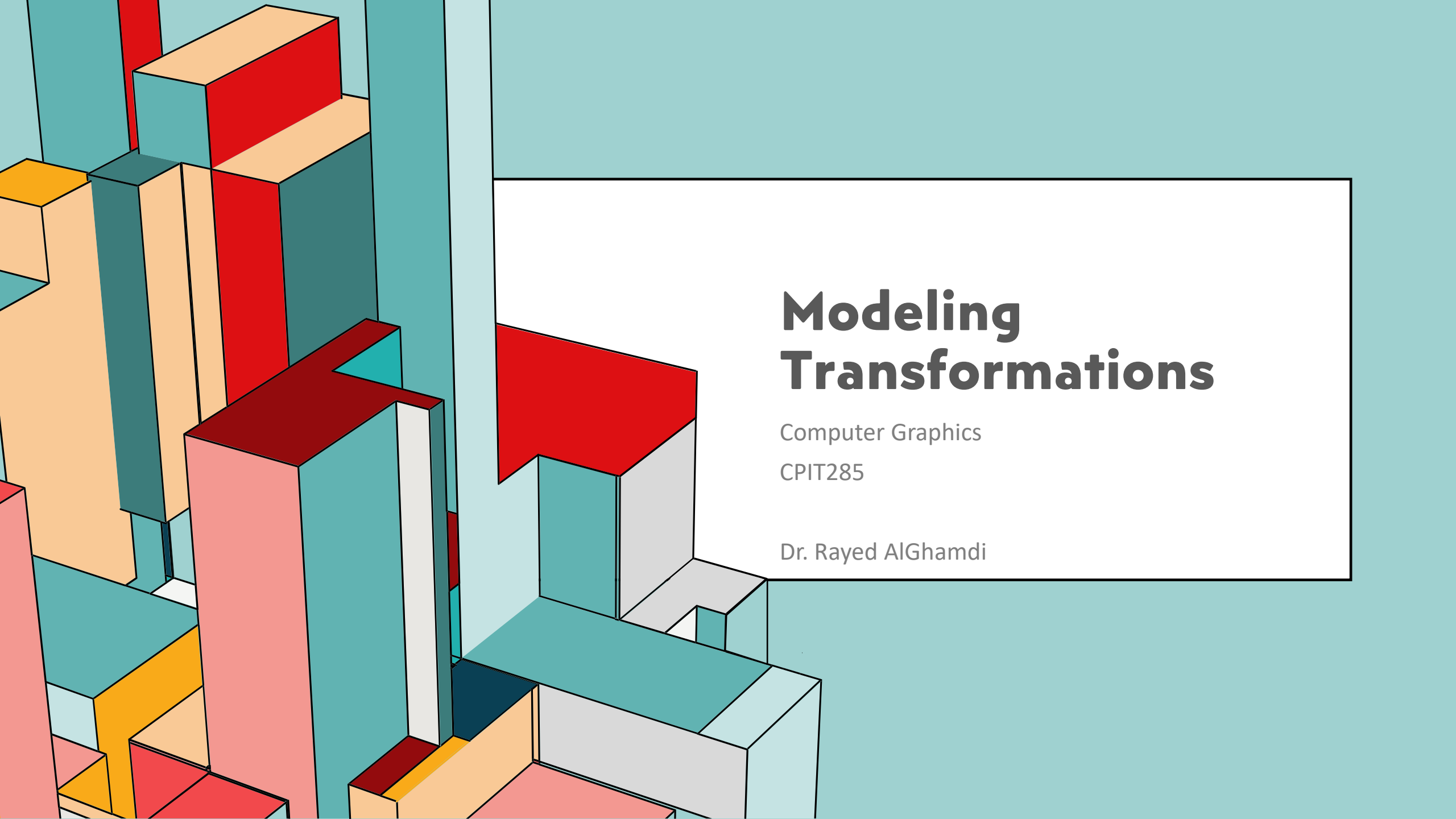




Modeling Transformations

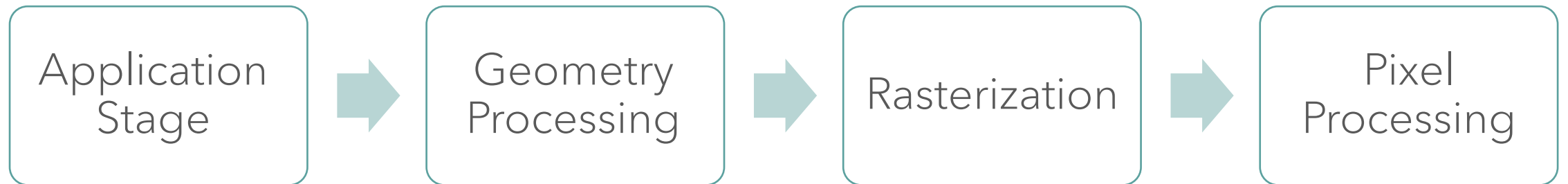
Computer Graphics

CPIT285

Dr. Rayed ALGhamdi

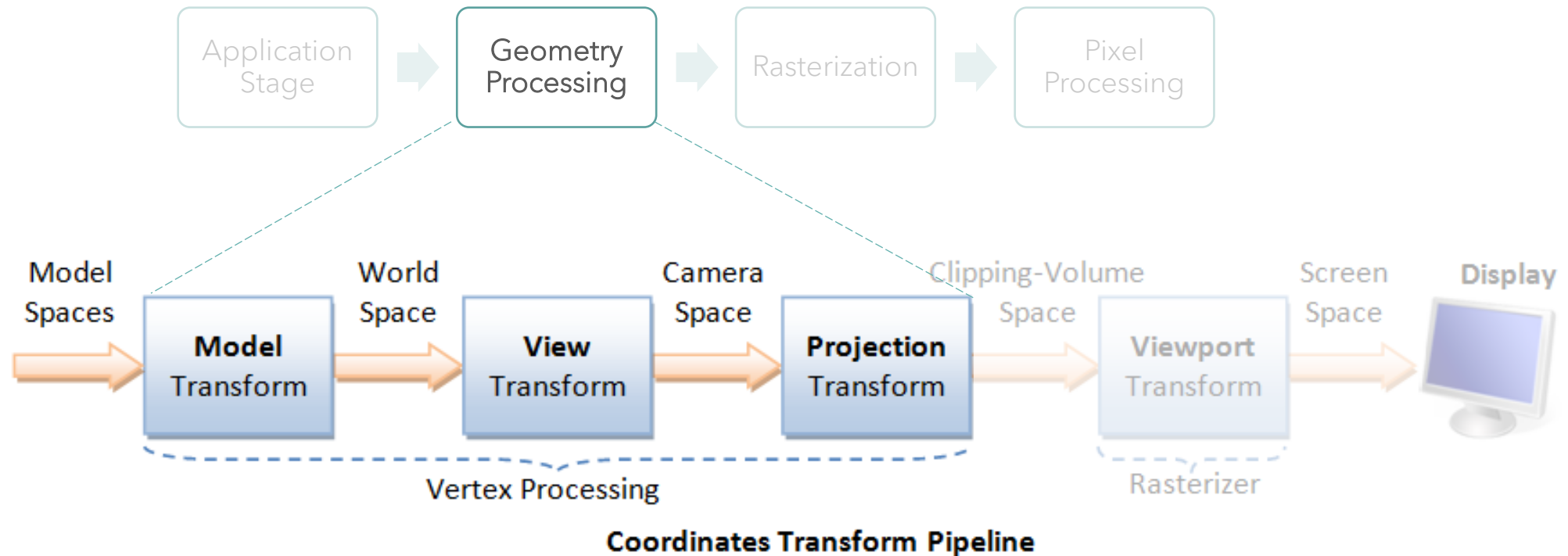
PREFACE

- Remember the graphics pipeline we studied in section 4?
- Highly recommended to review the content of that section before you proceed here.



PREFACE

- The geometry processing stage involve lots of processes in order to move to the next pipeline stage

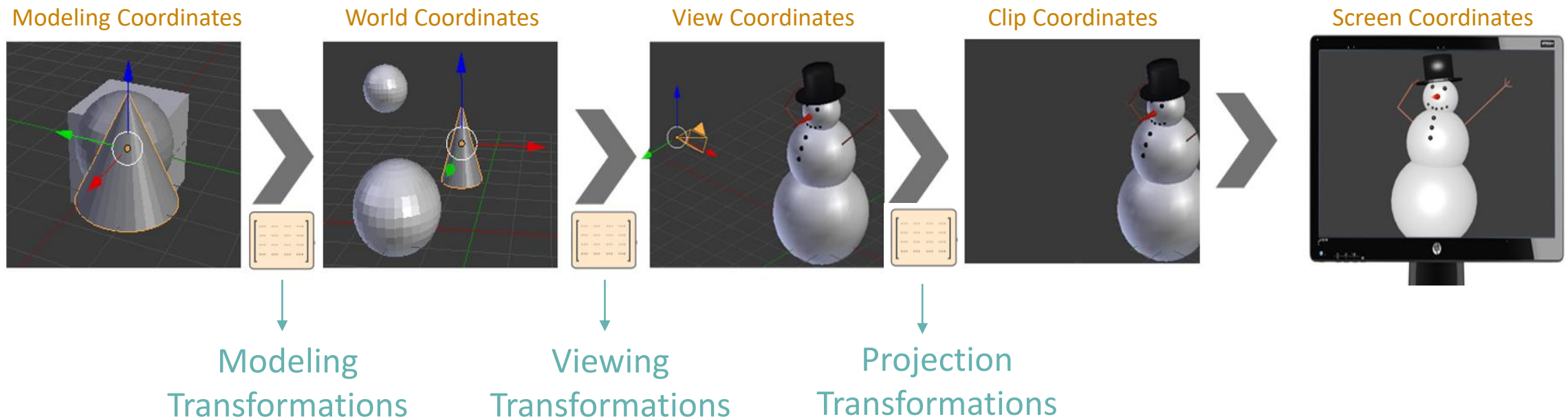


PREFACE

- So far, you realize how to manipulate objects on Blender.
- When an object (sphere, cube, cylinder...) is inserted, you change its original position, rotate it, resize it, then adjust the light settings and camera position to create the desired scene.
- Blender makes things so easy for you, it is a matter of drag and drop!
- As someone specialized in computing, you need to understand the underneath processes.
- That's why you will be involved in some of coding using OpenGL.
- While you are involved in the coding, you fully realize every stage.

PREFACE

- So easy to manipulate objects on GUI applications such as Blender. Transforming from one coordinate system to another is done without even realizing!
- When we code, you need to handle the processes of transformations...

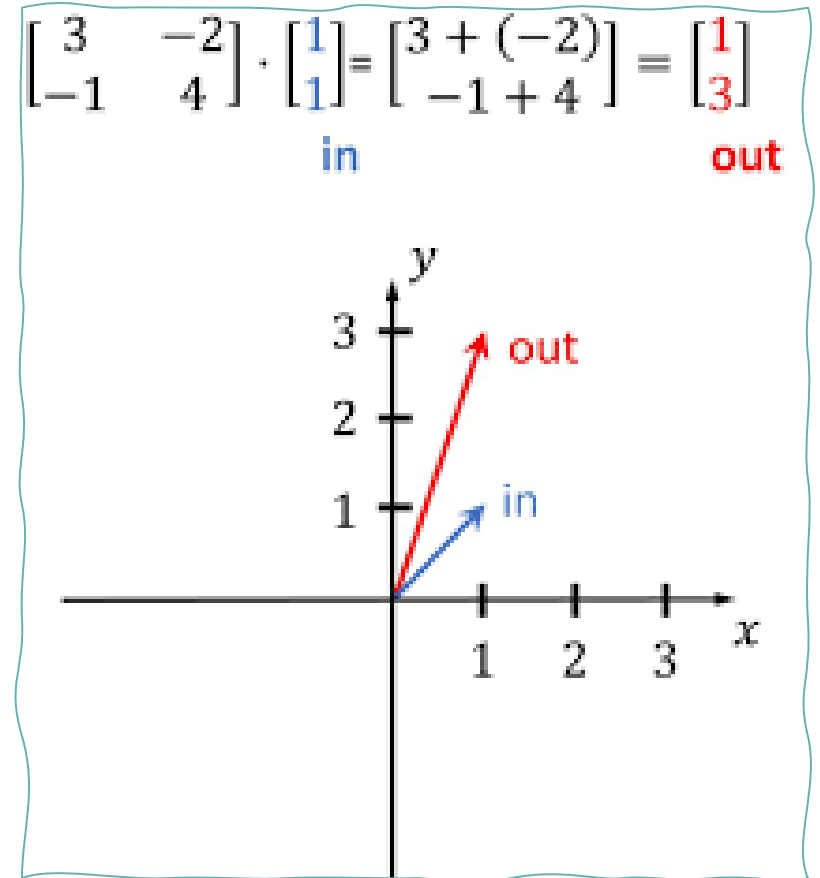


PREFACE

- When drawing objects, they are moved, rotated, and resized... these what it's called **Modeling Transformations**.
- The process relate to adjusting the view (camera direction) to create a scene is called **Viewing Transformations**.
- To calculate which object/part of an object nears the camera to look bigger and which is far to look smaller is called **Projection Transformation**.
- To understand and perform these transformations, you need to understand the process of **Matrices**.
- Matrices are very powerful mathematical constructs that seem scary at first, but once you'll grow accustomed to them, they'll prove extremely useful.

PREFACE

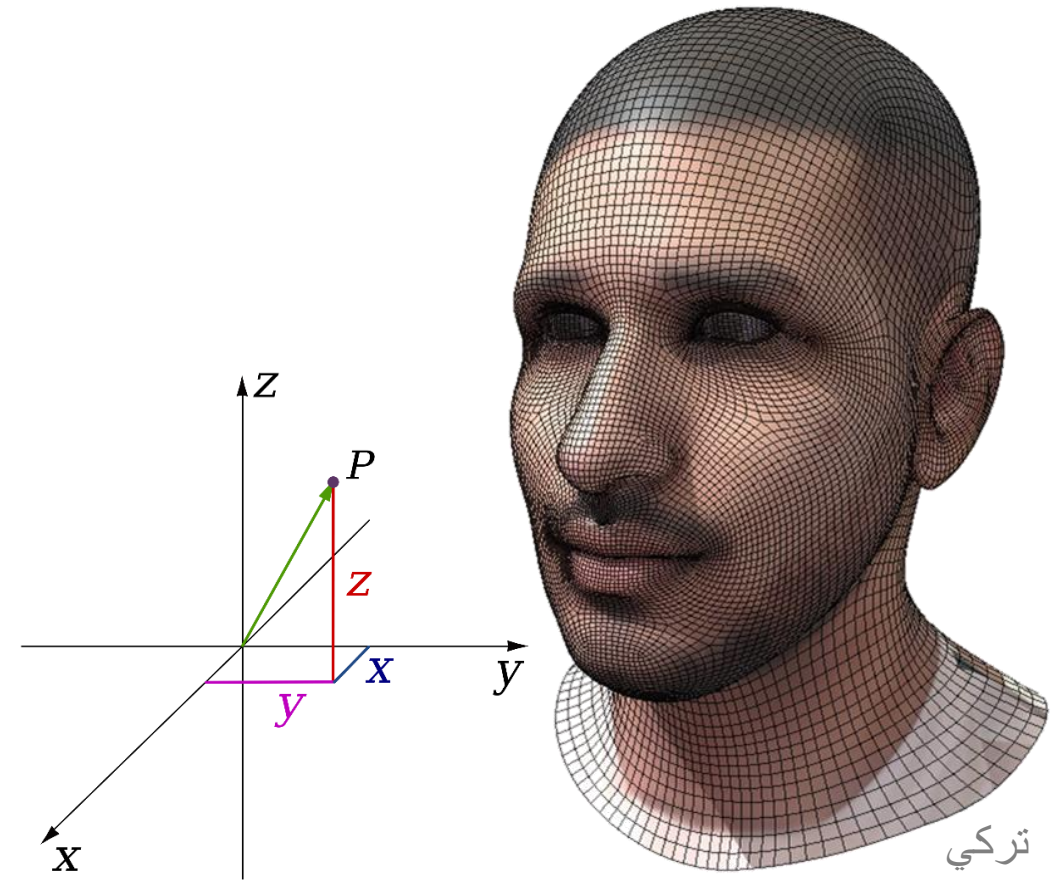
- To fully understand transformations, we first have to delve a bit deeper into some mathematical background, in specific [Linear Algebra](#).
- It is not essential that you know the mathematical details that are covered in this section, since they can be handled internally in OpenGL or by software libraries.
- However, you will need to be familiar with the concepts and the terminology.



LINEAR ALGEBRA

LINEAR ALGEBRA

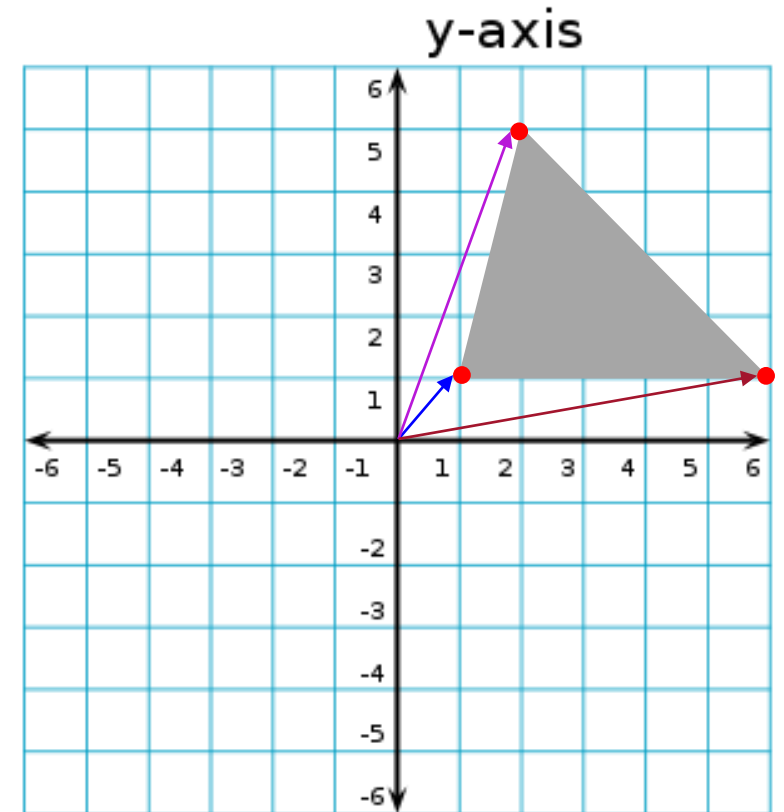
- Computer graphics heavily rely on Linear Algebra
- Linear Algebra is a branch of mathematics that is fundamental to computer graphics.
- It studies vectors, linear transformations, and matrices.
- Remember that a 3D model is seeded in vertices.
- A computer deal with each vertex as **vector**.



LINEAR ALGEBRA

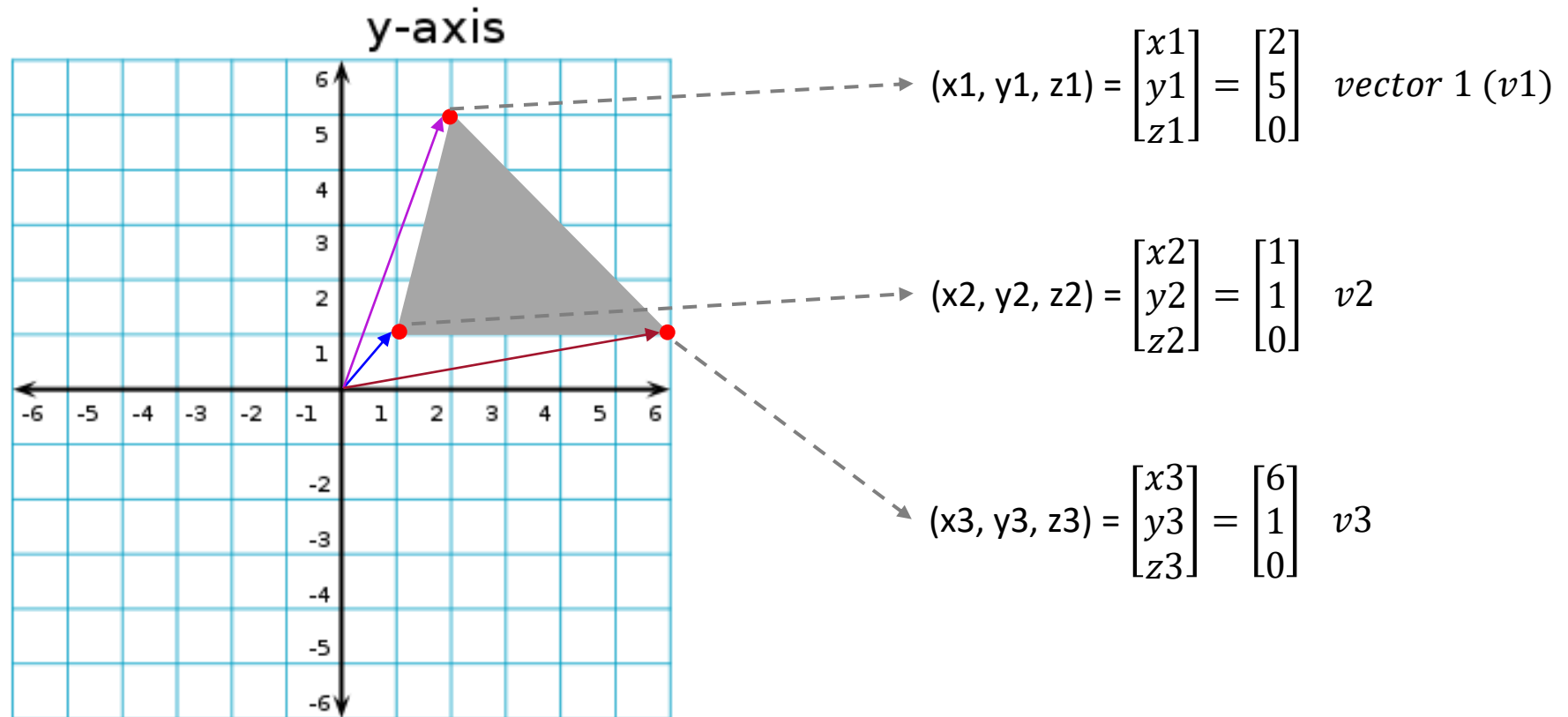
Vectors and Vector Math

- A vector is a quantity that has a length and a direction.
- A vector can be visualized as an arrow.
- How many vectors can be found in this simple drawing?
 - Three vectors
 - Each can be represented using x,y coordinates
 - When with 3D, then Z coordinate is added



LINEAR ALGEBRA

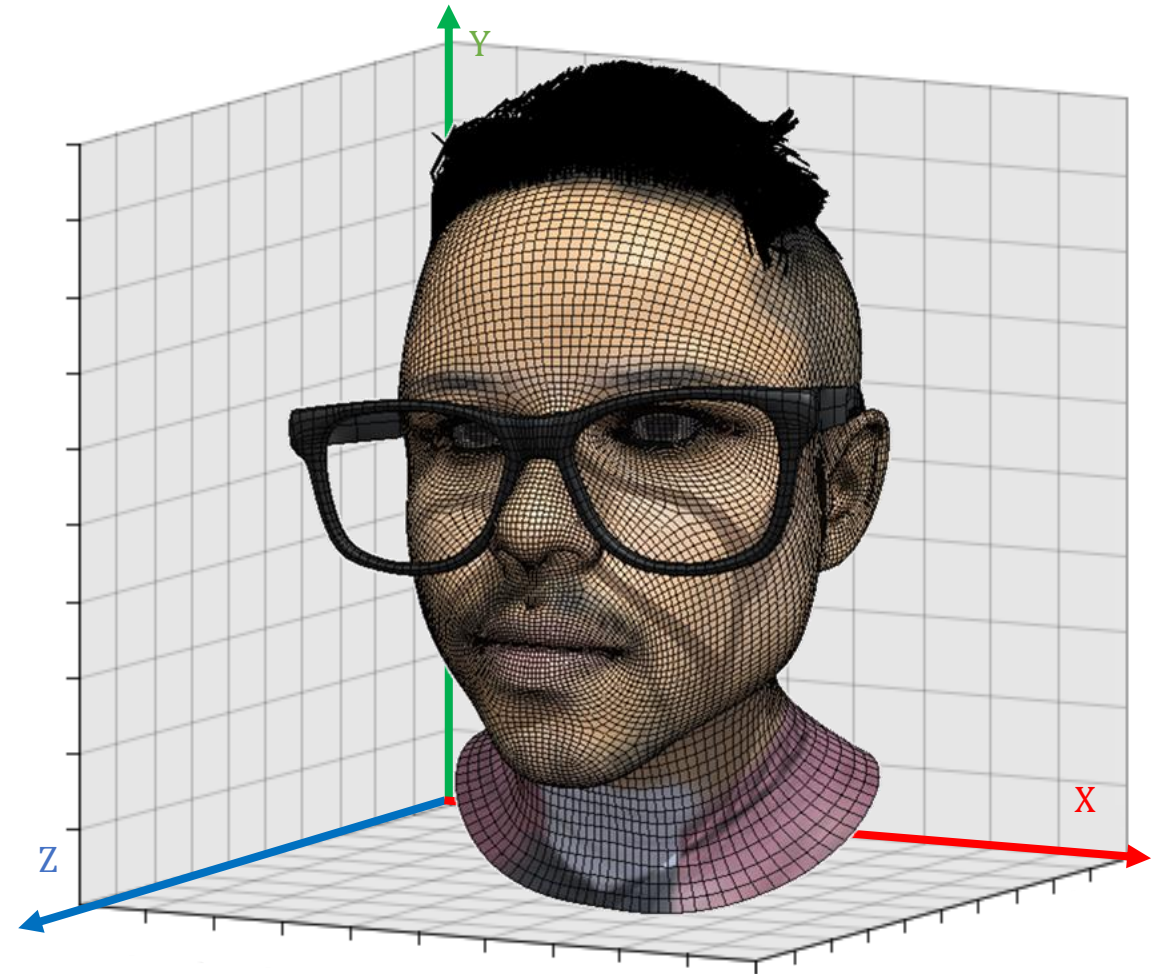
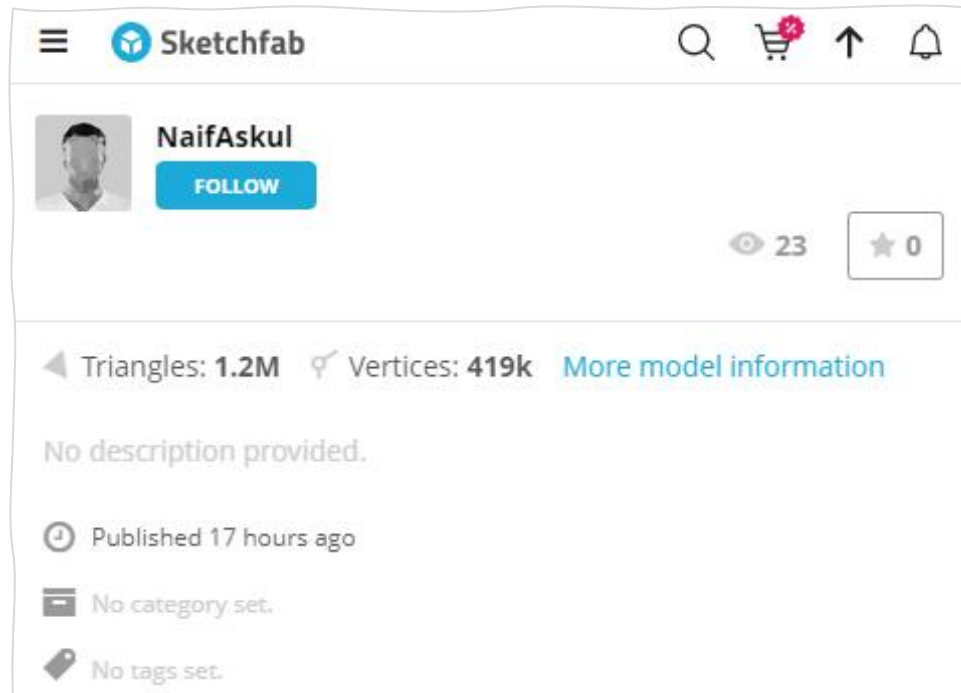
Vectors and Vector Math



LINEAR ALGEBRA

Vectors and Vector Math

How many vectors in Naif's head model?



LINEAR ALGEBRA

Matrices

- A matrix is a rectangular array of mathematical expressions, much like a two-dimensional array.
- Below is an example of a matrix displayed in the common square brackets form.

$$a = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$



Matrices values are indexed by (i,j) where i is the row and j is the column. That is why the matrix displayed here is called a 3-by-2 matrix. To refer to a specific value in the matrix, for example, 5, the a₃₁ notation is used.

LINEAR ALGEBRA

Matrices Basic Operations

To get a bit more familiar with the concept of an array of numbers, let's first look at a few basic operations.

○ Addition and subtraction

- Both matrices have to be exactly the same dimensions.
- like regular numbers, add/subtract each item to its equivalent in the second matrix.

$$\begin{bmatrix} 3 & 2 \\ 0 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 3+4 & 2+2 \\ 0+2 & 4+2 \end{bmatrix} = \begin{bmatrix} 7 & 4 \\ 2 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ 2 & 7 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 4-3 & 2-2 \\ 2-0 & 7-4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$$

LINEAR ALGEBRA

Matrices Basic Operations

○ Scalar Product

- It is also called: “dot product”. A scalar is a single number.
- The scalar product is a scalar multiplied by a matrix.
- It is as straightforward as addition and subtraction.
- The values in the matrices are each multiplied by the scalar.

$$2 \cdot \begin{bmatrix} 10 & 6 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 \cdot 10 & 2 \cdot 6 \\ 2 \cdot 4 & 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 20 & 12 \\ 8 & 6 \end{bmatrix}$$

LINEAR ALGEBRA

Matrices Basic Operations

○ Vector Product

- A matrix with one column is called a vector.
- To calculate the product of a matrix and a vector, the vector is written as a 4-by-1 matrix.
- The expressions to the right of the equals sign show how the new x, y and z values are calculated after the vector has been transformed.

$$\begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} a \cdot x + b \cdot y + c \cdot z + d \cdot 1 \\ e \cdot x + f \cdot y + g \cdot z + h \cdot 1 \\ i \cdot x + j \cdot y + k \cdot z + l \cdot 1 \\ m \cdot x + n \cdot y + o \cdot z + p \cdot 1 \end{pmatrix}$$

LINEAR ALGEBRA

Matrices Basic Operations

○ Vector Product

- Multiplying a vector by matrix produces a vector.

$$\begin{bmatrix} 2 & 3 & 7 & 4 \\ 1 & 6 & 3 & 4 \\ 2 & 4 & 5 & 2 \\ 6 & 5 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 4 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 2x2 + 3x4 + 7x3 + 4x2 \\ 1x2 + 6x4 + 3x3 + 4x2 \\ 2x2 + 4x4 + 5x3 + 2x2 \\ 6x2 + 5x4 + 2x3 + 3x2 \end{bmatrix} = \begin{bmatrix} 4 + 12 + 21 + 8 \\ 2 + 24 + 9 + 8 \\ 4 + 16 + 15 + 4 \\ 12 + 20 + 6 + 6 \end{bmatrix} = \begin{bmatrix} 45 \\ 43 \\ 39 \\ 44 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 4 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1x4 + 2x5 + 1x3 \\ 4x4 + 3x5 + 2x3 \\ 1x4 + 2x5 + 3x3 \end{bmatrix} = \begin{bmatrix} 17 \\ 37 \\ 23 \end{bmatrix}$$

LINEAR ALGEBRA

Matrices Basic Operations

○ Matrix-Matrix Product

- The product of a matrix with another matrix is quite a bit more complex.
- In order to multiply two matrices, the number of columns in the first matrix must be equivalent to the number of rows in the second matrix.
- The order of the output matrix takes the number of rows in the first matrix and the number of columns in the second matrix.

$$\begin{bmatrix} 2 & 1 & 4 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 6 & 3 & -1 & 0 \\ 1 & 1 & 0 & 4 \\ -2 & 5 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 27 & -2 & 12 \\ -1 & 6 & 0 & 6 \end{bmatrix}$$

2×3 3×4 2×4

match
result

$$\begin{bmatrix} 5 & 6 & 1 \\ 0 & 5 & 7 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} -3 & -1 & 0 \\ -2 & 8 & 1 \end{bmatrix}$$

3×3 2×3

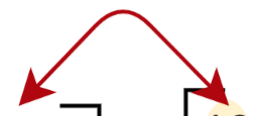
do not match

undefined
answer

LINEAR ALGEBRA

Matrices Basic Operations

○ Matrix-Matrix Product


$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \end{bmatrix}$$
$$= \begin{bmatrix} 1 \times 10 + 2 \times 20 + 3 \times 30 & 1 \times 11 + 2 \times 21 + 3 \times 31 \\ 4 \times 10 + 5 \times 20 + 6 \times 30 & 4 \times 11 + 5 \times 21 + 6 \times 31 \end{bmatrix}$$
$$= \begin{bmatrix} 10+40+90 & 11+42+93 \\ 40+100+180 & 44+105+186 \end{bmatrix} = \begin{bmatrix} 140 & 146 \\ 320 & 335 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1 \cdot 3 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot 1 \\ 2 \cdot 3 + 1 \cdot 0 & 2 \cdot 0 + 1 \cdot 1 \end{bmatrix}$$
$$= \begin{bmatrix} 3 & 0 \\ 6 & 1 \end{bmatrix}$$

LINEAR ALGEBRA

Matrices Basic Operations

○ Matrix-Matrix Product

- Find out the product

$$\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix} = ?$$

$$\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 5 & 6 \end{bmatrix} = ?$$

- What is the matrix dimension of the product

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} =$$

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} =$$

LINEAR ALGEBRA

Matrices Basic Operations

○ The Identity Matrix

- The Identity Matrix is a special kind of matrix that is similar to the concept of “1” in real numbers. Just like a real number multiplied by “1” results in the real number itself, any matrix multiplied by the identity matrix results in the matrix itself.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot x + 0 \cdot y + 0 \cdot z + 0 \cdot 1 \\ 0 \cdot x + 1 \cdot y + 0 \cdot z + 0 \cdot 1 \\ 0 \cdot x + 0 \cdot y + 1 \cdot z + 0 \cdot 1 \\ 0 \cdot x + 0 \cdot y + 0 \cdot z + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot x \\ 1 \cdot y \\ 1 \cdot z \\ 1 \cdot 1 \end{pmatrix}$$

LINEAR ALGEBRA

Matrices Basic Operations

$$v' = v \cdot M$$

$$\begin{matrix} M \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix} \cdot \begin{matrix} v \\ \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \end{matrix} = \begin{pmatrix} 1 \cdot x + 0 \cdot y + 0 \cdot z + 0 \cdot 1 \\ 0 \cdot x + 1 \cdot y + 0 \cdot z + 0 \cdot 1 \\ 0 \cdot x + 0 \cdot y + 1 \cdot z + 0 \cdot 1 \\ 0 \cdot x + 0 \cdot y + 0 \cdot z + 1 \cdot 1 \end{pmatrix} = \begin{matrix} v' \\ \begin{pmatrix} 1 \cdot x \\ 1 \cdot y \\ 1 \cdot z \\ 1 \cdot 1 \end{pmatrix} \end{matrix}$$

identity matrix

- v → pre-transformation coordinates
- v' → transformation coordinates
- So far we're working in 2D - (X,Y) coordinates - but OpenGL treats this as 3D coordinates with Z=0

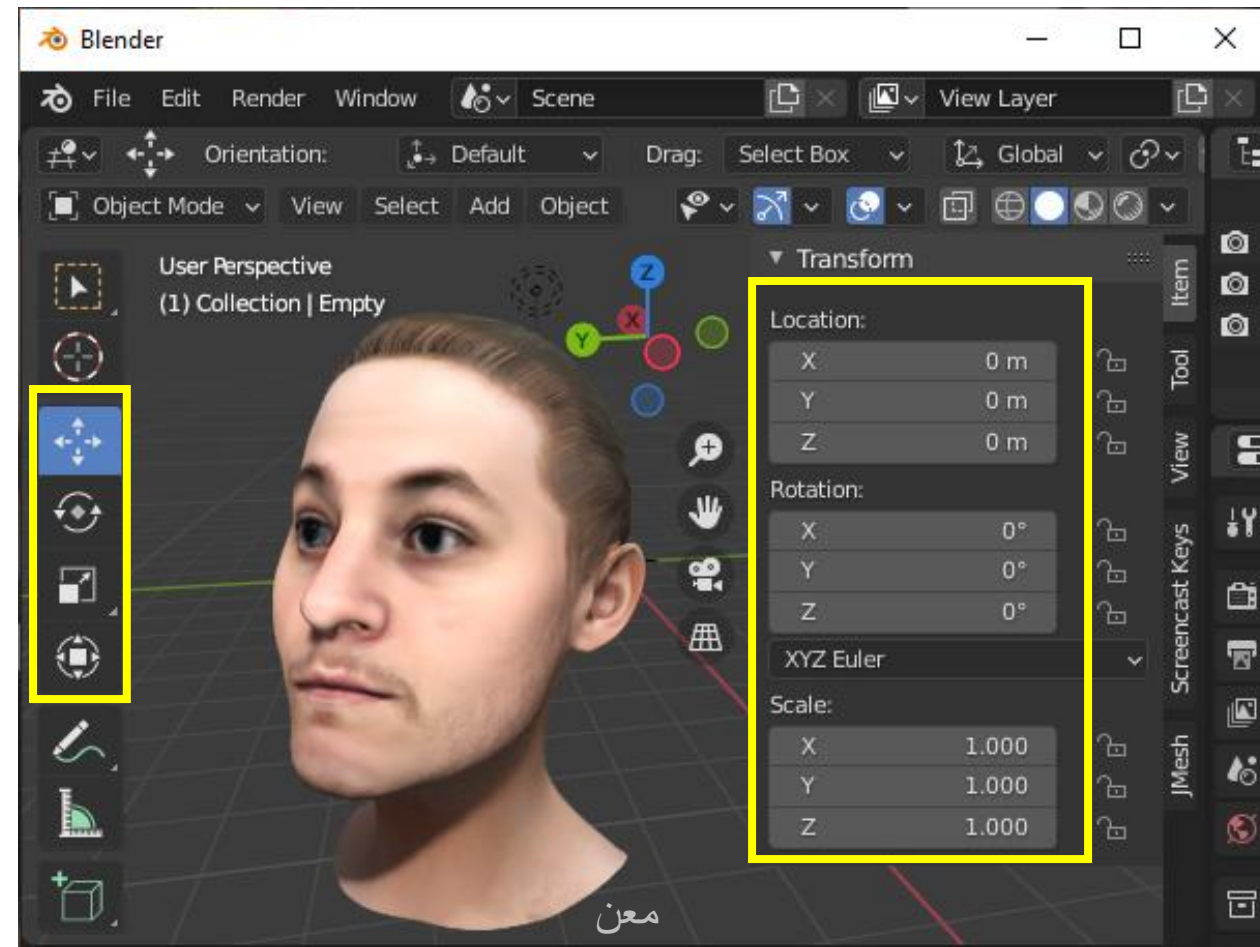
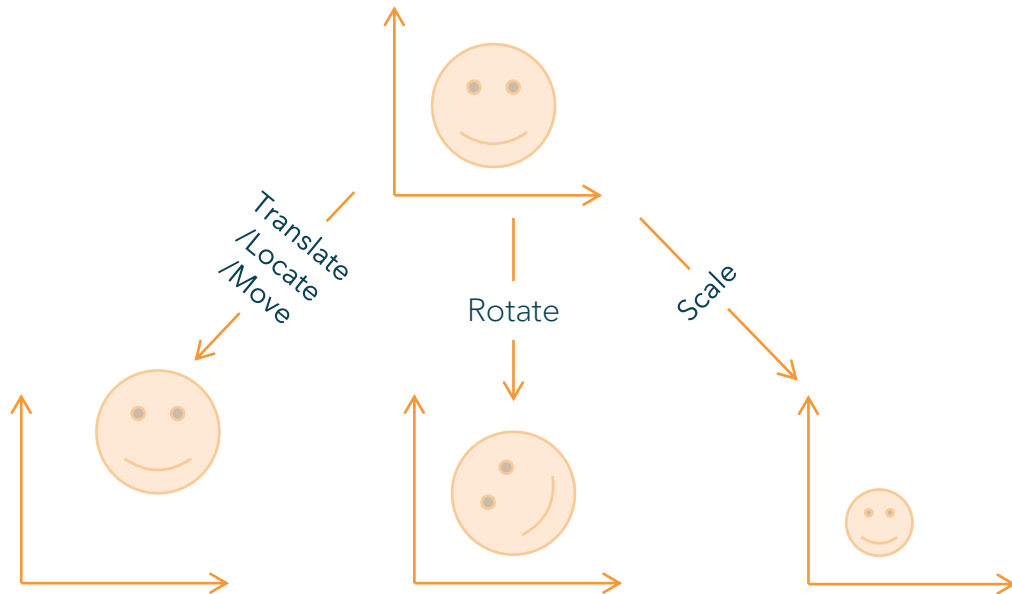
Modeling

Transformations

MODELING TRANSFORMATIONS

Blender Transforming Options

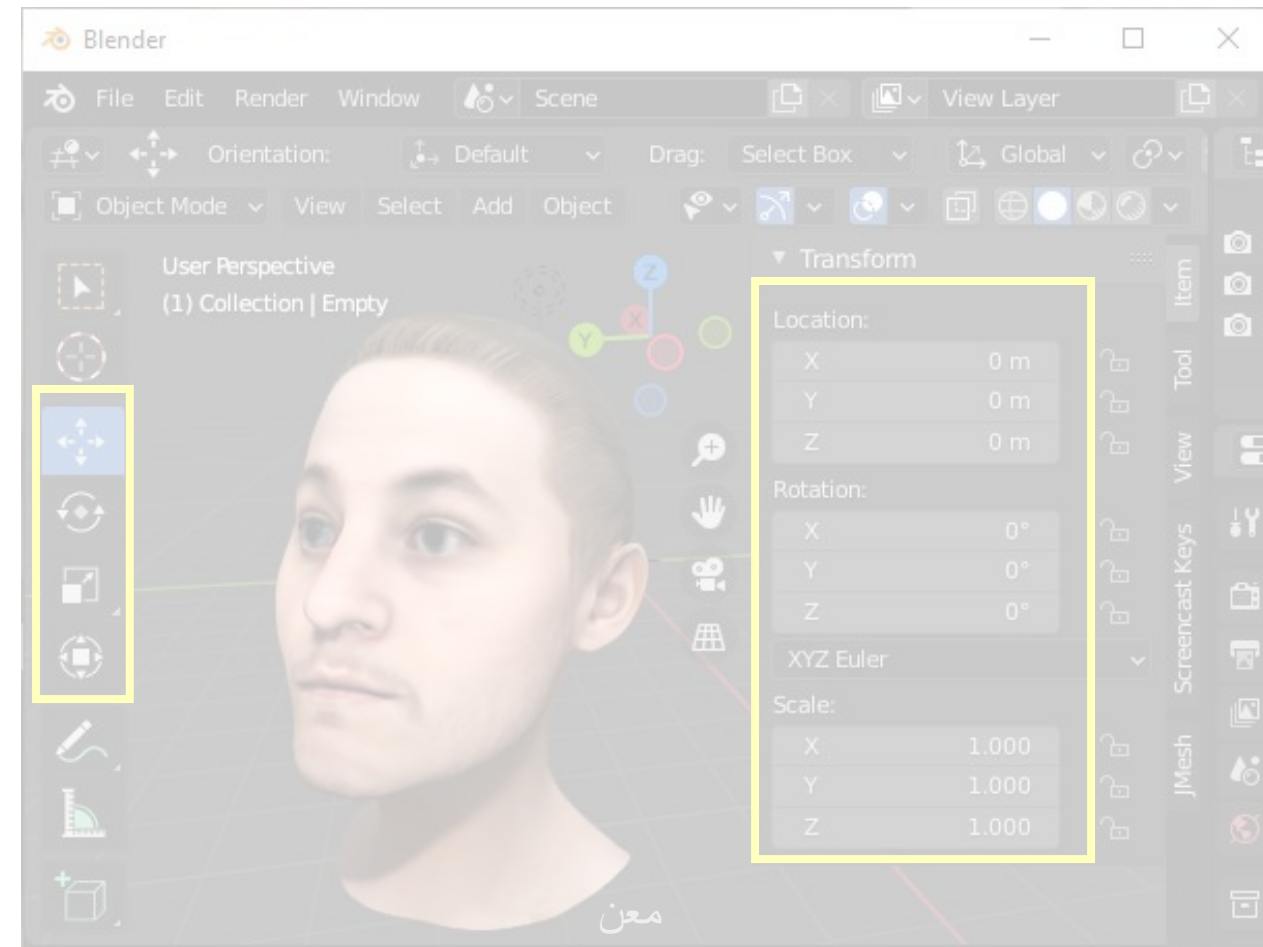
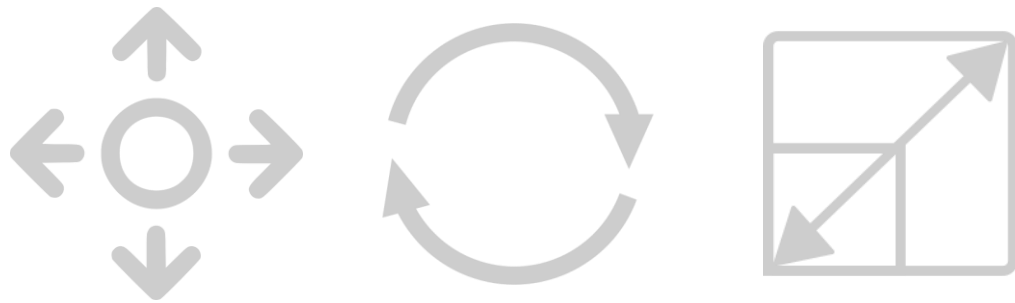
- What modeling transformation did you play with when modeling your head on Blender?



MODELING TRANSFORMATIONS

Blender Transforming Options

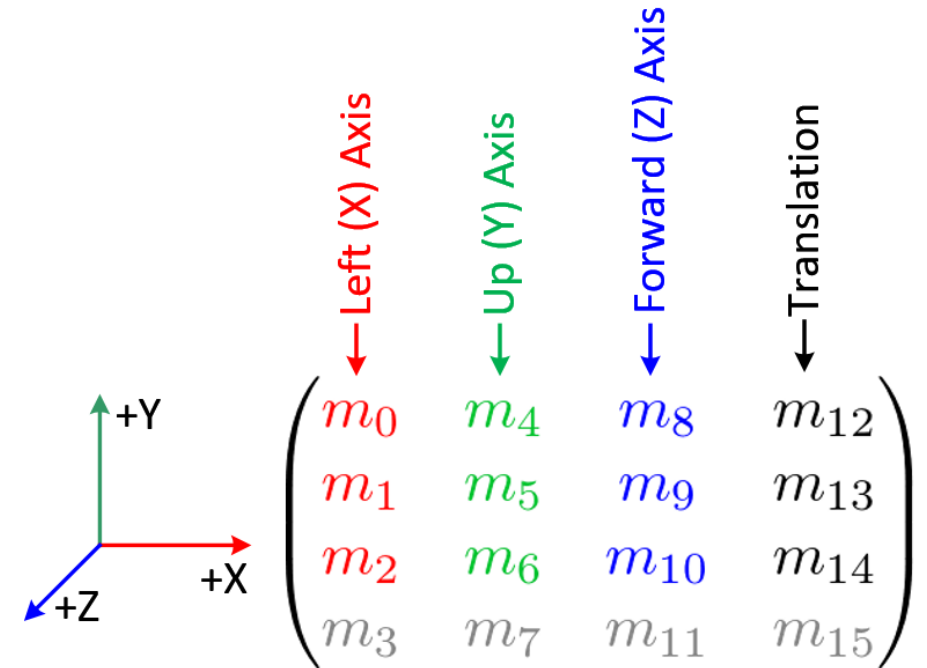
- How do you perform these transforming options?
 - It is a matter of clicking and using the mouse/keyboard!



MODELING TRANSFORMATIONS

OpenGL Transforming Options

- Using OpenGL to code your graphical application, you use the same transformation options: move, rotate and scale.
- How do you do that in coding?
- To conduct the modeling transformations, OpenGL uses 4x4 matrix and then multiply the matrix with each axis value (x, y, z)



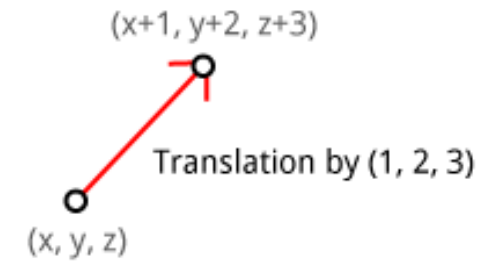


TRANSLATE

MODELING TRANSFORMATIONS

Translation

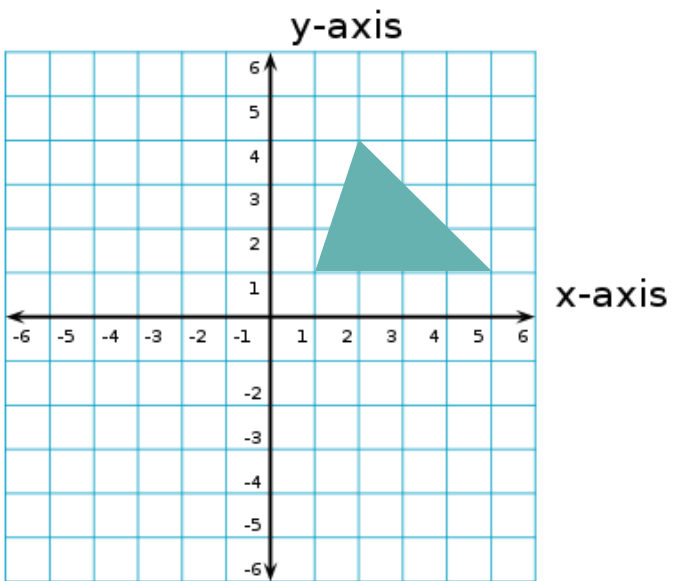
- To move an object from one position to another; the addition process takes place to the matrix by adding the value of the distance to which an object is to be moved.
- The process applies to all vertices of the object.



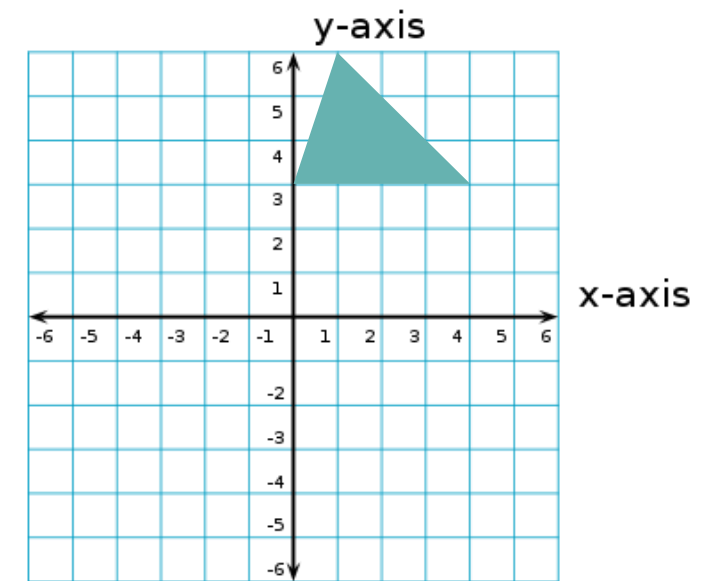
$$\begin{bmatrix} 1 & 0 & 0 & Tx \\ 0 & 1 & 0 & Ty \\ 0 & 0 & 1 & Tz \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x + Tx \\ y + Ty \\ z + Tz \\ 1 \end{pmatrix}$$

MODELING TRANSFORMATIONS

Translation



```
glTranslatef(-1.0, 2.0, 0.0);
```



GL_TRIANGLES

V1 = (1.0, 1.0, 0.0)

V2 = (2.0, 4.0, 0.0)

V3 = (5.0, 1.0, 0.0)

GL_TRIANGLES

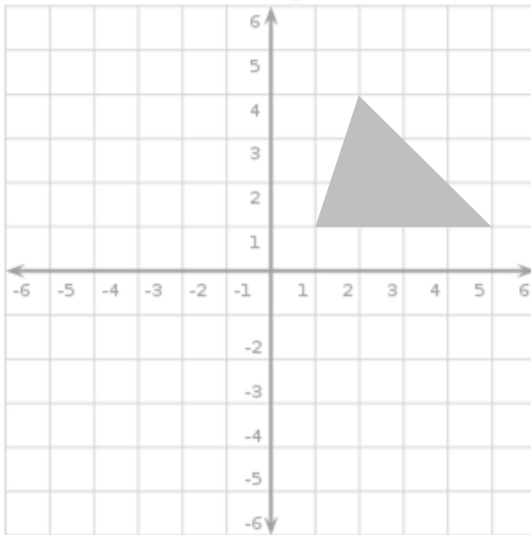
V1' = (0.0, 3.0, 0.0)

V2' = (1.0, 6.0, 0.0)

V3' = (4.0, 3.0, 0.0)

MODELING TRANSFORMATIONS

Translation



GL_TRIANGLES

V1 = (1.0, 1.0, 0.0)

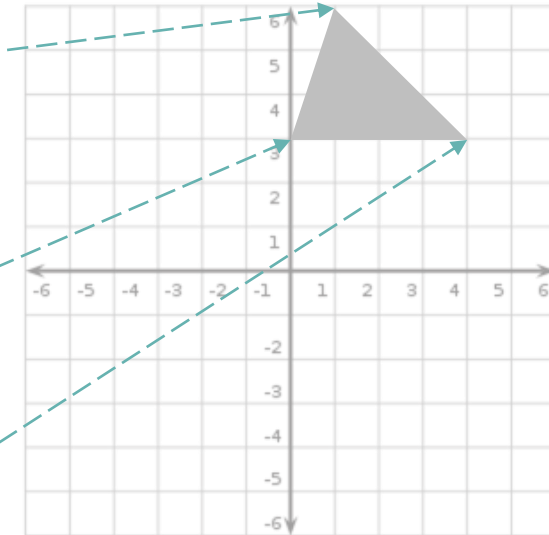
V2 = (2.0, 4.0, 0.0)

V3 = (5.0, 1.0, 0.0)

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 4 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 + (-1) \\ 4 + 2 \\ 0 + 0 \\ 0 + 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + (-1) \\ 1 + 2 \\ 0 + 0 \\ 0 + 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 + (-1) \\ 1 + 2 \\ 0 + 0 \\ 0 + 1 \end{bmatrix}$$



GL_TRIANGLES

V1' = (0.0, 3.0, 0.0)

V2' = (1.0, 6.0, 0.0)

V3' = (4.0, 3.0, 0.0)

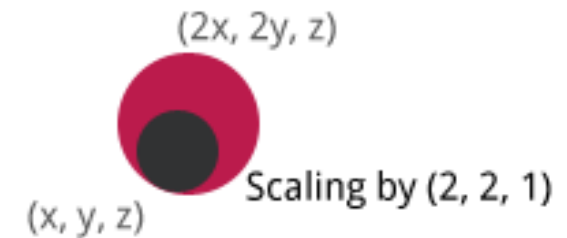


SCALING

MODELING TRANSFORMATIONS

Scaling

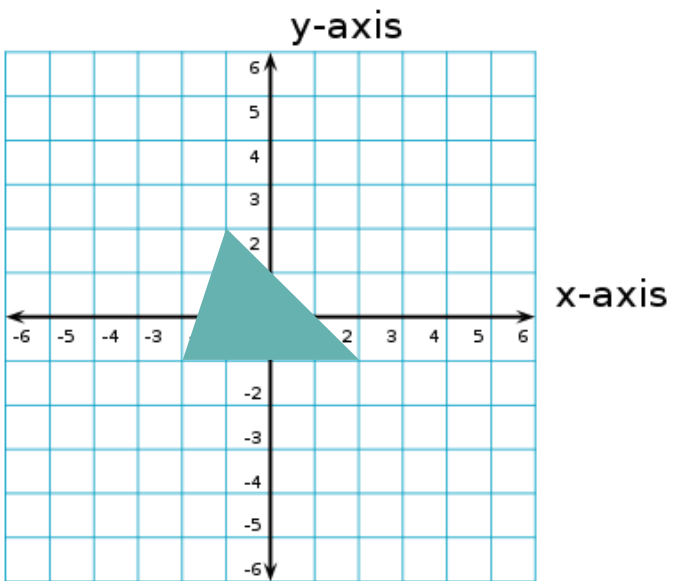
- To resize an object using the 4x4 matrix, values of the current coordinates are multiplied by the new sizing value.
- The process applies to all vertices of the object.



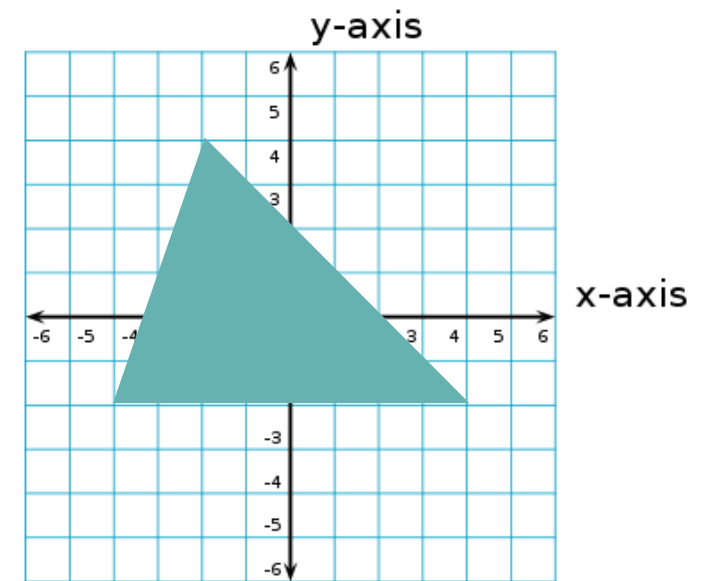
$$\begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \cdot S_x \\ y \cdot S_y \\ z \cdot S_z \\ 1 \end{pmatrix}$$

MODELING TRANSFORMATIONS

Scaling



```
glScalef(2.0, 2.0, 0.0);
```



GL_TRIANGLES

V1 = (-2 , -1 , 0)

V2 = (-1 , 2 , 0)

V3 = (2 , -1 , 0)

GL_TRIANGLES

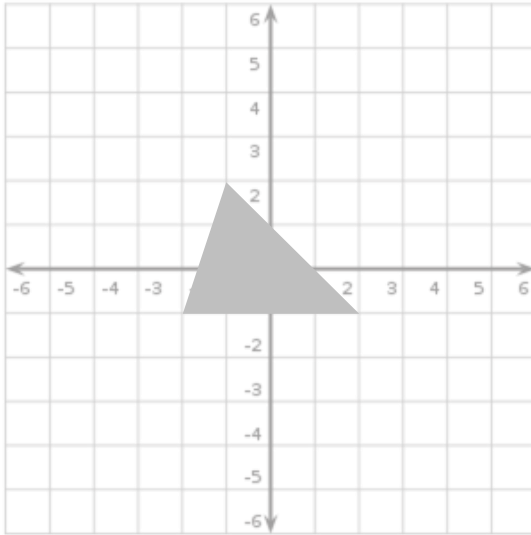
V1' = (-4, -2, 0)

V2' = (-2, 4, 0)

V3' = (4, -2, 0)

MODELING TRANSFORMATIONS

Scaling



GL_TRIANGLES

V1 = (-2 , -1 , 0)

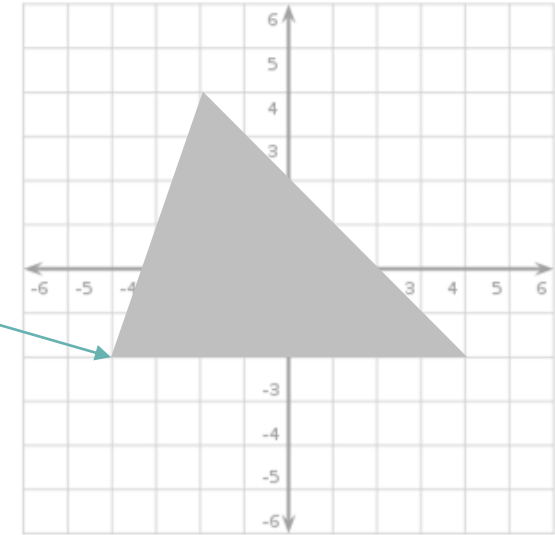
V2 = (-1 , 2 , 0)

V3 = (2 , -1 , 0)

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 * (-2) \\ 2 * (-1) \\ 1 * 0 \\ 1 * 1 \end{bmatrix}$$

...

...



GL_TRIANGLES

V1' = (-4, -2, 0)

V2' = (-2, 4, 0)

V3' = (4, -2, 0)



ROTATION

MODELING TRANSFORMATIONS

Rotation

- A rotation transformation rotates a vector around the origin (0,0,0) using a given axis and angle.
- Objects can be rotated around any given axis.
- The matrices for rotating around the three axes are specified here.
- The rotation angle is indicated by the theta (ϑ).

Rotation around X-axis:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ \cos \theta \cdot y - \sin \theta \cdot z \\ \sin \theta \cdot y + \cos \theta \cdot z \\ 1 \end{pmatrix}$$

Rotation around Y-axis:

$$\begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} \cos \theta \cdot x + \sin \theta \cdot z \\ y \\ -\sin \theta \cdot x + \cos \theta \cdot z \\ 1 \end{pmatrix}$$

Rotation around Z-axis:

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} \cos \theta \cdot x - \sin \theta \cdot y \\ \sin \theta \cdot x + \cos \theta \cdot y \\ z \\ 1 \end{pmatrix}$$

MODELING TRANSFORMATIONS

Rotation

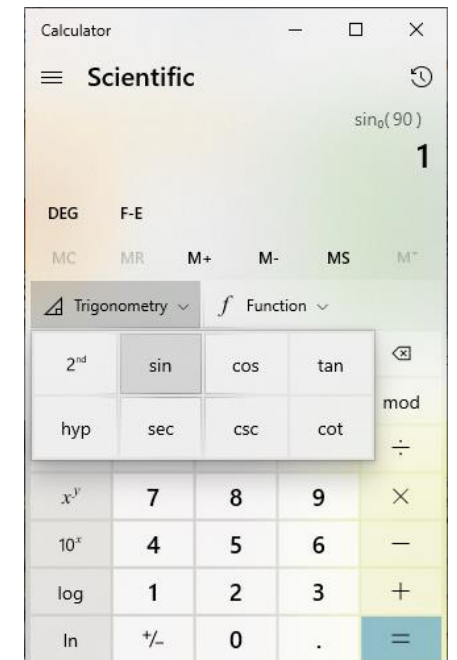
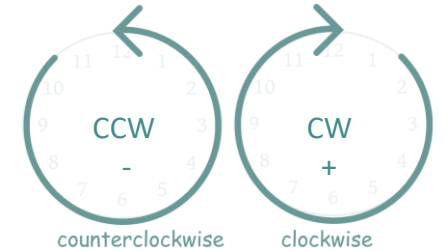
- How is the matrix of rotation produced?
 - Identify the angle and direction of rotation (CW > positive angle value, CCW > negative angle value)
 - Calculate the sin & cos of the given angle
 - Determine on which axis do you want to rotate and fill the values as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

rotate around X-axis

rotate around Y-axis

rotate around Z-axis



MODELING TRANSFORMATIONS

Rotation

- Example1: Rotate around Y axis by 180 degree

- $\sin(180) = 0 \mid \cos(180) = -1$

$$\begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation
matrix around
Y-axis by 180

- Example2: Rotate around X axis by -35 degree

- $\sin(-35) = -0.57 \mid \cos(-35) = 0.81$

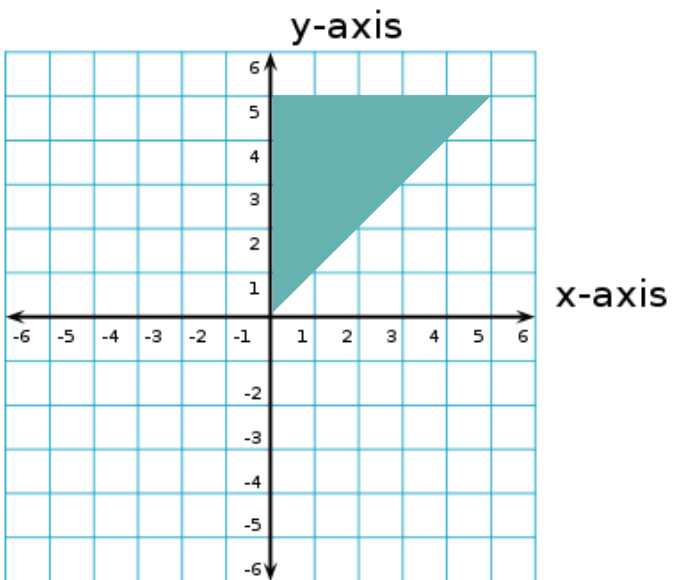
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.81 & 0.57 & 0 \\ 0 & -0.57 & 0.81 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation
matrix around
X-axis by -35

MODELING TRANSFORMATIONS

Rotation



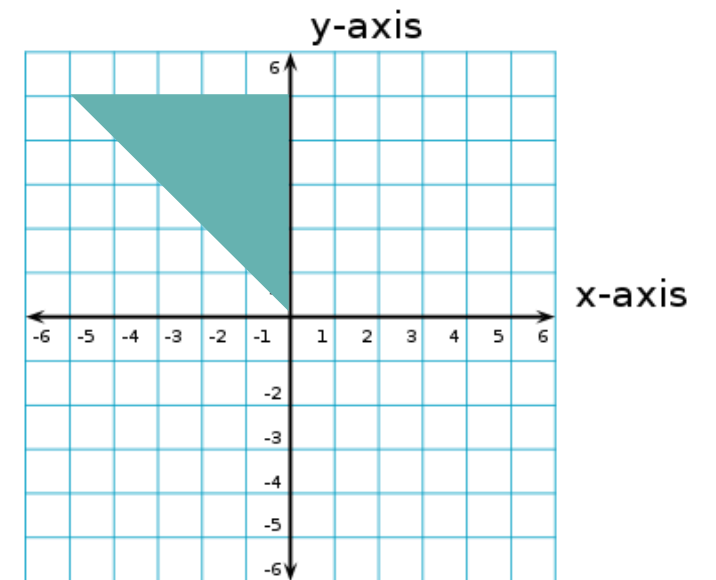
GL_TRIANGLES

V1 = (0, 0, 0)

V2 = (0, 5, 0)

V3 = (5, 5, 0)

```
glRotatef(180.0, 0.0, 1.0, 0.0);
```



GL_TRIANGLES

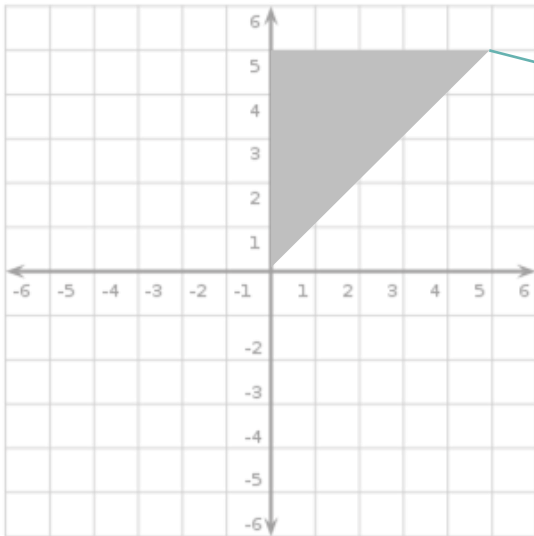
V1 = (0, 0, 0)

V2 = (0, 5, 0)

V3 = (-5, 5, 0)

MODELING TRANSFORMATIONS

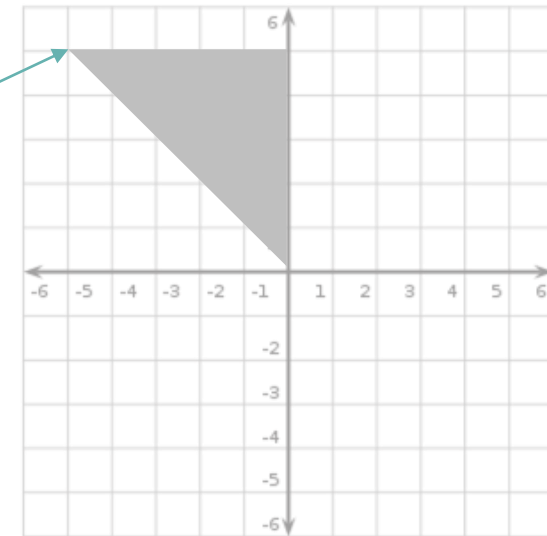
Rotation



$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 5 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 * 5 \\ 1 * 5 \\ -1 * 0 \\ 1 * 1 \end{bmatrix}$$

...

...



GL_TRIANGLES

V1 = (0, 0, 0)

V2 = (0, 5, 0)

V3 = (5, 5, 0)

GL_TRIANGLES

V1 = (0, 0, 0)

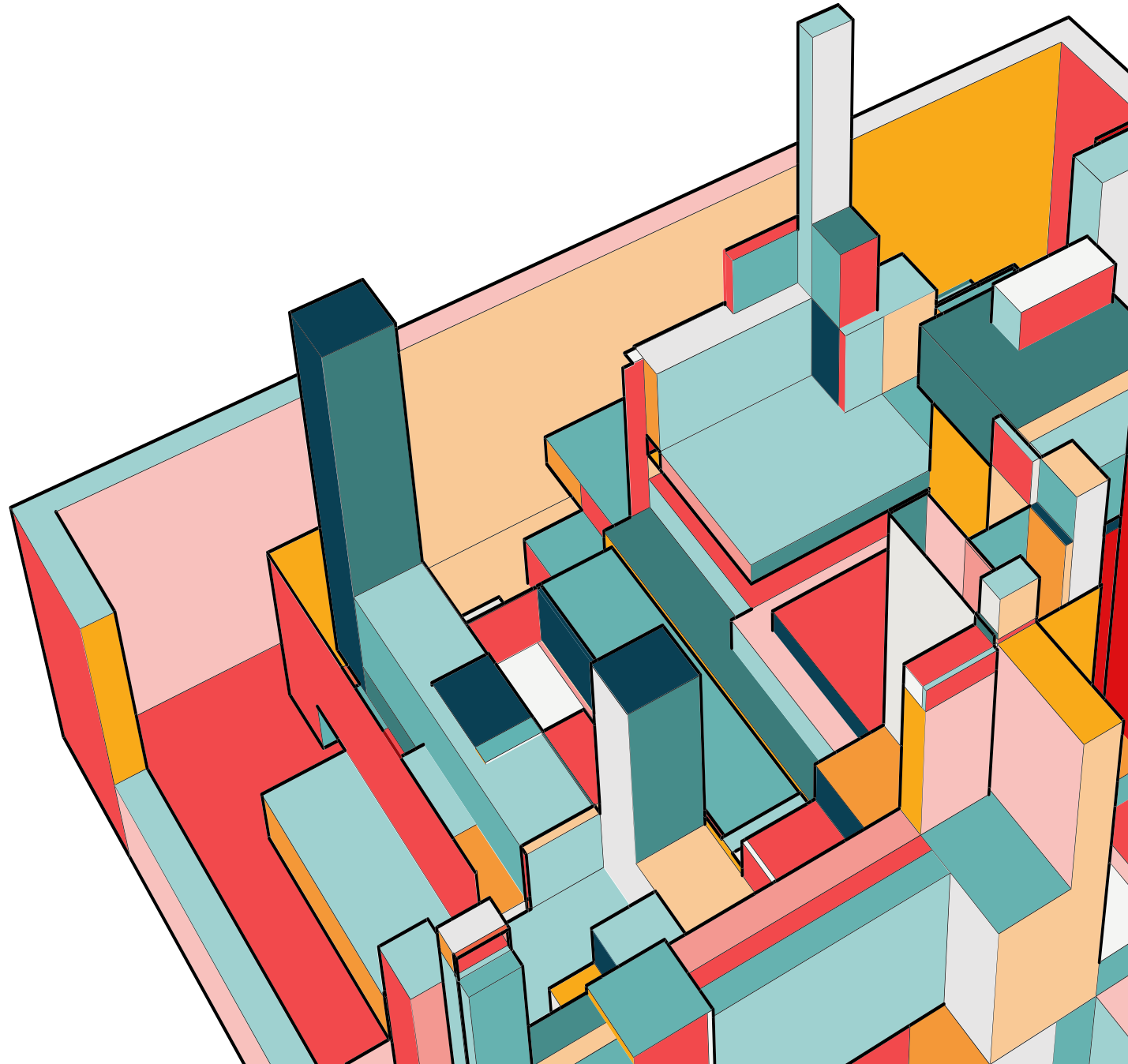
V2 = (0, 5, 0)

V3 = (-5, 5, 0)

100% CONFIDENT

Your journey in this course is
enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

Dr. Rayed ALGhamdi

+966-12-640-0000 ex:75560

raalghamdi8@kau.edu.sa

www.raalghamdi8.kau.edu.sa

